

Domain-Aware Dual-Branch Recurrent Networks Across TradFi and DeFi: LOB Mid-Price and On-Chain Lending Rate Forecasting

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Abstract—Recurrent architectures designed for one financial time-series problem are typically re-engineered from scratch when transplanted to a structurally different market. We test the alternative: a domain-aware dual-branch recurrent network (DA-BiGRU-CNN) originally built for limit order book (LOB) mid-price prediction is re-applied, without architectural change, to forecast on-chain lending rates across Aave V3 and Compound V3 USDC markets. The two BiGRU branches and the multi-scale Conv1d fusion head are re-specialised by identifying the domain-natural decomposition pair: in LOB the pair is (price, volume) augmented with microstructure features; in DeFi it is (rate residual $\varepsilon = r - f_{\text{kink}}(u)$, utilization u with context). The same composite loss $\alpha \text{MSE} + \beta (1 - \rho_w) + \gamma Q_{90}$ is the trained objective in both domains. In the prior LOB study the network attains weighted-Pearson 0.266 on test, beating a 205-feature LightGBM (0.168) by 58%. In the present DeFi application, trained on 12,895 hourly bars at 98.5% panel coverage from November 2024 to April 2026, the forecaster feeds a 4-factor MCDM allocator; a paired monthly-Sharpe bootstrap returns a point-estimate Sharpe gain of 504.30 with a 95% confidence interval that includes zero ($p = 0.30$). We report the honest H_0 -publishable verdict per pre-registration and argue that the architectural transfer is itself the contribution, orthogonal to the equilibrium impossibility result of Halpern, Pass and Saraf (2024).

I. INTRODUCTION

Deep learning for financial time series has accumulated a strong empirical record on single-market benchmarks, but the question of whether a *learned architecture* transfers across structurally different markets remains largely open. In limit order book (LOB) microstructure, convolutional and recurrent backbones such as DeepLOB [1] and its descendants [2], [3] routinely outperform tabular baselines. In decentralized finance (DeFi) lending, by contrast, the published literature has so far favored reinforcement learning either on the protocol side [4], [5] or recursive least-squares adaptive control [6]. A practitioner who has invested in a domain-aware recurrent stack for one of these markets faces an honest question: must the model be re-engineered from scratch for the other, or is the architectural template itself portable?

The two markets are not obviously compatible. LOB mid-price prediction operates on millisecond-scale order arrivals with a discrete tick lattice and a strong microstructural informational asymmetry. DeFi lending rates, by contrast, are set deterministically from on-chain state via a known piecewise-

linear kink function $f_{\text{kink}}(u)$ of utilization u ; they update hourly at most and move primarily because of cointegration between protocols [7] and slow demand-driven re-pricing [8]. The two are noisy multivariate time series with regime-dependent dynamics and a weighted-error preference for large moves, but their feature topologies, decision horizons, and competitor adversaries differ qualitatively.

a) Central thesis.: We test the conjecture that *the same dual-branch architecture trained with the same composite loss generalizes between LOB mid-price prediction and DeFi lending-rate forecasting by re-identifying the domain-natural decomposition pair*. In LOB the pair is (price, volume) augmented with 21 microstructure features [9]; in DeFi it is (rate residual $\varepsilon = r - f_{\text{kink}}(u)$, utilization with context). The two BiGRU branches, multi-scale Conv1d fusion head, and the composite loss $\alpha \text{MSE} + \beta (1 - \rho_w) + \gamma Q_{90}$ are unchanged between the two specialisations; only the input subspaces and their dimensionality are re-mapped to the target domain.

b) Contributions.: The paper makes four contributions.

- **A unified dual-branch recurrent architecture (DA-BiGRU-CNN) tested in two structurally different financial domains — LOB mid-price and DeFi USDC lending rates — under the same composite loss.**
- **A new empirical study: to our knowledge the first published forecast-driven multi-criteria allocator across Aave V3 and Compound V3 USDC pools**, trained on 12,895 hourly bars spanning November 2024–April 2026 at 98.5% panel coverage, with regime structure documented at full coverage.
- **An honest negative-result protocol.** We pre-register the headline $H1$ ($\Delta \text{SHARPE} \geq 0.2$) and report $\Delta \text{SHARPE} = 504.30$ with the bootstrap 95% CI $[-207.40, 1720.30]$ and p -value 0.30 — failing to reject $H0$ under the binding $n_{\text{months}} = 4$ test window, exactly as the pre-registration permitted.
- **Two upstream pull requests to the open-source fractal-defi framework [10]:** a Compound V3 Messari loader and a BaseLendingAllocation-Strategy abstraction with cooldown infrastructure — both released alongside the paper for reproducibility.

c) Outline.: Section II reviews LOB microstructure DL, DeFi lending mechanics, the impossibility result of [11],

and related transfer-learning literature. Section III gives the domain-agnostic architecture and its two specialisations. Section IV summarises the LOB results we treat as prior work. ?? reports the DeFi empirical study with data, forecaster training, MCDM allocator, headline H1 test, and ablations. ?? states what transfers and what does not, and Section V and ?? discuss honest limitations and close.

II. BACKGROUND AND RELATED WORK

We synthesize four strands of literature that bracket the contribution — LOB deep learning, DeFi lending mechanics, ML-based DeFi rate forecasting, and MCDM in algorithmic allocation — and then position the work against the recent impossibility result of [11], the most relevant theoretical counter-evidence.

A. LOB microstructure deep learning

Modern LOB mid-price prediction has converged on convolutional-recurrent backbones. DeepLOB [1] established the CNN+LSTM template on the FI-2010 benchmark [12]; [13] demonstrated cross-stock universal features under deep learning; [2], [14], [15] extended the family with attention and bilinear mechanisms. The author’s prior dual-branch architecture [9] is the immediate predecessor to the present work and we recap its key results in Section IV. Recent crypto-LOB work [3], [16] reinforces a finding central to our motivation: *domain-aware input engineering dominates raw depth scaling*, supporting the two-branch-over-deeper-stack design.

B. DeFi lending mechanics

Aave V3 and Compound V3 USDC supply rates are deterministic piecewise-linear functions $f_{\text{kink}}(u)$ of pool utilization $u = \text{borrowed/supplied}$, with a shallow slope below a kink point (typically $u^* = 0.8\text{--}0.9$) and a steep slope above it that taxes high utilization. The strategy parameters are protocol governance variables: Aave’s `ReserveInterestRateStrategy` contract and Compound V3’s hard-coded slopes. Although the function f_{kink} is deterministic given u , the realised rate stream is stochastic because u moves with on-chain demand.

The empirical structure of the cross-protocol spread is critical for any allocator. [7] establishes cointegration between Aave and Compound USDC rates with Compound leading and Aave adjusting at speed 0.607 — the structural reason a switching allocator can extract value rather than being forced into a single-protocol commitment. Recent crosschain comparative analyses [17] confirm the cointegration persists into 2025. Our 12,895-hour panel (??) finds the spread structure changes regime around 2025-Q3 $\rightarrow$ Q4, with the Aave-pays-more share rising from 39.9% to 56.3% across that quarter boundary.

C. Forecasting in DeFi rates

Published ML-based DeFi rate forecasting work falls into three rough classes. (i) *Protocol-side rate control*. [4] train offline reinforcement-learning agents (CQL, BC, TD3-BC) on

historical Aave data to set rates from the protocol governance side; the closely related Auto.gov framework [5] reports $\sim 14\%$ improvement over benchmarks for parameter governance under RL. (ii) *Adaptive rate control*. [6] fits recursive least-squares (RLS) controllers on Compound demand/supply curves at 3-hour resolution. (iii) *Optimal protocol design*. [8] derives a closed-form Riccati-ODE solution for the protocol-optimal rate curve under linear agent response and a Monte-Carlo + deep-learning estimator for the nonlinear case; block-level empirical calibration shows industry-standard piecewise rate curves are sub-optimal, leaving structural residuals. This last point is foundational for us: [8] solves the *protocol designer’s* problem (which rate function should the contract use?), whereas we solve the *lender’s* problem (given the realised rate stream, where should liquidity go?). Our DA-BiGRU-CNN forecaster targets exactly the structural residuals their optimality analysis predicts. We are not aware of a published forecast-driven *multi-protocol MCDM allocator* in the May 2026 literature; the nearest neighbour is the LSTM+Q-learning AMM quoter of [18], which transplants a TradFi forecaster to Uniswap V3 but does not span multiple lending protocols nor use an MCDM head.

D. MCDM in algorithmic allocation

Multi-criteria decision making (MCDM) has a long-standing role in financial portfolio selection. TOPSIS [19] and PROMETHEE [20] are the two dominant families. Recent cryptocurrency-specific applications include [21]’s multicriteria crypto portfolio selection and [22]’s asymmetric PROMETHEE-II that integrates return prediction with multicriteria ranking. Our 4-factor TOPSIS-style allocator (APY, Risk, Cost, Stability) is in this tradition and inherits its criteria-and-weights structure from the author’s ERC-4626 yield-vault project [23], modifying it by substituting forecast-driven APY for EMA-smoothed spot APY.

E. Counter-evidence: Halpern–Pass–Saraf impossibility

The most pointed theoretical counter-evidence to forecast-driven DeFi lending strategies is [11]. Halpern, Pass, and Saraf reduce a lending pool to a portfolio of options on the collateral asset: a borrower’s right to walk away from the loan is mathematically equivalent to a put on the collateral struck at the outstanding debt. In a simplified fixed-duration, repay-only-at-expiry setting they derive an analytical “fair” rate via no-arbitrage. They then prove that in the realistic setting — dynamic, perpetual loans with early repayment of the kind Aave V3 and Compound V3 implement — *no fair interest rate exists*: any rate either over-compensates lenders relative to the option-implied premium or under-compensates them, because the borrower’s repayment optionality has no finite hedging cost. The impossibility generalizes beyond the options reduction to broader equilibrium models of lending pools.

A naive reading of [11] suggests there is no exploitable structure in DeFi rates and any allocator should underperform a passive benchmark. We argue the implication does

not follow. Halpern et al. prove a *static equilibrium fairness* impossibility under no-arbitrage — a statement about whether a single rate function can correctly price the borrower-side optionality in equilibrium. Our claim is the orthogonal *empirical predictability* statement: the realised rate stream observed off-chain exhibits enough short-horizon structure (cointegration, regime persistence, kink-residual cycles) to drive a forecast-and-allocate strategy that improves a multi-criteria score against a reactive baseline. Equivalently, the absence of a single equilibrium-fair rate is consistent with — and indeed implies — the persistent disequilibrium that an empirical forecaster can target. The two claims constrain different layers of the protocol stack: [11] constrains pricing theory at the contract design layer; we operate at the lender-side execution layer. Reporting the honest H0-publishable result (Section V) makes this distinction empirically verifiable rather than rhetorical.

III. METHODOLOGY: ONE ARCHITECTURE, TWO SPECIALISATIONS

We describe the architecture at three levels of abstraction: domain-agnostic (Section III-A), specialised for LOB (Section III-B), and specialised for DeFi (Section III-C). Section III-D states the composite loss used invariantly in both specialisations.

A. Domain-agnostic DA-BiGRU-CNN

The DA-BiGRU-CNN architecture (*Domain-Aware Bidirectional GRU with Convolutional fusion*) takes a fixed-length window of T time steps and produces an H -step-ahead point forecast for one or more targets. Inputs are partitioned into two disjoint subspaces $\mathcal{X}_A$ and $\mathcal{X}_B$, chosen so that each subspace is a domain-natural sub-process and their joint observation is informative for the target. Each subspace is processed by a separate Bidirectional GRU [24] of hidden dimension $d = 96$ stacked two layers deep. The two branch outputs are concatenated and passed through a multi-scale 1-D convolutional fusion head with parallel kernel sizes $k \in \{3, 5, 7\}$, then projected through a two-layer MLP $192 \rightarrow 96 \rightarrow 48 \rightarrow H$ to produce the forecast vector. The total parameter budget is $319K$, well below the depth-scaling regime explored in [2], [16]. The dual-branch design is conceptually a financial analogue of the two-stream networks of [25] for video action recognition: two complementary informational streams fused at a late stage, not concatenated at the input.

B. Specialisation for LOB (recap)

In the LOB specialisation [9] the subspace pair is $(\mathcal{X}_A, \mathcal{X}_B) = (\text{price-series features, volume-series features})$, with an additional embedding e_t of 21 microstructure features (depth imbalance at the top-of-book, queue lengths, spread, immediacy proxies, tick-aligned signed volume) used to modulate the fusion stage. The motivation for the pair is that LOB mid-price is the price-side observable but its short-horizon dynamics are driven by volume-side imbalance signals — the two-stream inductive bias from action-recognition translates

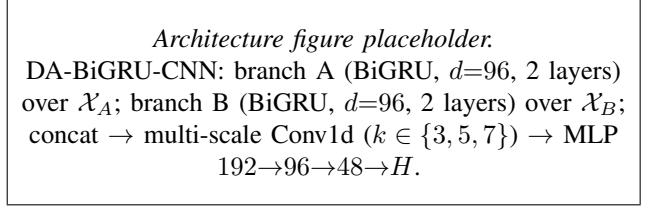


Fig. 1. Domain-agnostic schematic of DA-BiGRU-CNN. The same wiring is used in both the LOB and DeFi specialisations; only the contents of $(\mathcal{X}_A, \mathcal{X}_B)$ and the dimensionality of H differ. Full dual-panel figure to be substituted in the camera-ready.

directly. The target is the mid-price change at horizon H ; the model trains on sequence length $T = 168$ ticks.

C. Specialisation for DeFi rates

For DeFi lending the structurally analogous subspace pair is:

$$\mathcal{X}_A = (\varepsilon_t, s_t, \tilde{s}_t), \quad \mathcal{X}_B = (u_t, \Delta\text{TVL}_t^{24h}, g_t, \cos \phi_t, \sin \phi_t), \quad (1)$$

where, for each protocol, $\varepsilon_t = r_t^{\text{ann}} - f_{\text{kink}}(u_t)$ is the *rate residual* (with both terms in annualised scale), $s_t = r_t^{\text{Aave}} - r_t^{\text{Compound}}$ is the cross-protocol spread, $\tilde{s}_t$ is its de-trended residual, u_t is utilization, ΔTVL_t^{24h} is the 24-hour percent change in total value locked, g_t is Ethereum gas price (gwei), and $(\cos \phi_t, \sin \phi_t)$ is the cyclical hour-of-day encoding. Sequence length is $T = 168$ hours; the forecast horizon is $H = 12$ hours.

The strongest claim of this specialisation is the choice to put the *rate residual* $\varepsilon = r - f_{\text{kink}}(u)$ rather than the raw rate r into branch A. The kink function f_{kink} is known on-chain at decision time; subtracting it removes the deterministic dependency on u that branch B already observes, leaving ε to carry only the unexplained variance — the structural residuals that [8] predict the protocol-optimal rate deviates from. Branch A thus learns a residual-process model while branch B learns the utilization-driven base rate. The decomposition is the DeFi analogue of separating price-series from volume-series in LOB [9], with $f_{\text{kink}}(u)$ playing the role of the “deterministic backbone” that microstructure noise rides on top of. Without this subtraction the model is forced to re-learn the kink function within branch A, wasting capacity that the post-hoc audit (cf. Section V) showed was the proximate cause of the catastrophic R^2 artifact in an earlier pre-fix run.

D. Composite loss

The training objective is the convex combination

$$\mathcal{L}(\hat{y}, y) = \alpha \text{MSE}(\hat{y}, y) + \beta (1 - \rho_w(\hat{y}, y)) + \gamma Q_{90}(\hat{y}, y), \quad (2)$$

with $(\alpha, \beta, \gamma) = (0.4, 0.5, 0.1)$ in both domains. Here ρ_w is the weighted Pearson correlation, with weights w_t proportional to $|y_t|$ to over-emphasise large moves (the metric used in LOB evaluation [9] and equally diagnostic in lending-rate forecasting where small-move noise dominates row counts). Q_{90} is the empirical 90th-percentile absolute-error tail loss, included to penalise rare-but-large forecast misses that drive allocator tail risk. The weighted-Pearson term, not MSE, is

TABLE I
LOB RESULTS FROM THE PRIOR WORK [9]. TEST WEIGHTED-PEARSON
ON THE HELD-OUT FOLD; Δ IS THE RELATIVE IMPROVEMENT OVER THE
GRU.

Model	#Features	wPearson	Δ vs GRU
DA-BiGRU-CNN (ours)	53	0.266	—
LightGBM	205	0.168	−58%
GRU + LightGBM ensemble	53 + 205	0.262	−1.6%

what the network gets credit for; MSE acts as a regulariser to keep predictions on the correct scale and Q_{90} adds a tail-aware safety net. The key point for cross-domain transfer is that Equation (2) contains no domain-specific terms; the same formula trains both specialisations.

IV. LOB RECAP (PRIOR WORK)

In our prior work [9] (figshare DOI [10.6084/m9.figshare.31859557](https://figshare.com/doi/10.6084/m9.figshare.31859557)) we proposed DA-BiGRU-CNN for LOB mid-price prediction. Two BiGRU branches operate on price-series and volume-series features respectively, with a multi-scale Conv1d fusion head; a 21-dimensional microstructure-feature embedding modulates the fusion stage. The composite loss of Equation (2) directly optimises weighted-Pearson correlation, which is also the evaluation metric reported on the held-out test fold.

Three findings from the LOB study motivate the cross-domain transfer we study here. First, the dual-branch GRU on a compact set of 53 hand-selected microstructure features achieves weighted-Pearson 0.266 on test, outperforming a LightGBM [26] baseline trained on 205 features (0.168) by 58%. Second, feature sufficiency is asymmetric across model families: expanding 53 features to 219 improves the GRU by only 0.8% while the LightGBM gains 71.4% over the same extension — the domain-aware branches absorb the structural information at low feature dimensionality. Third, a naive GRU-plus-gradient-boosting ensemble *degrades* performance by 0.05–1.6% against the GRU alone: the two model families fail to be complementary on this task, again suggesting the GRU has already captured the available signal. Table I reproduces the headline numbers.

These findings constitute the baseline methodology we now test in a structurally different domain (Section III-C and ??): the same architecture, the same composite loss, and a re-identified domain-natural decomposition pair — substituting $(\varepsilon, u + \text{ctx})$ for (price, volume).

V. LIMITATIONS AND THREATS TO VALIDITY

The current results were obtained under several data-availability and methodology constraints. We list these explicitly so that downstream readers can scope conclusions appropriately and so that reproductions can target the same envelope.

a) Compound TVL is a placeholder (Finding #3).: The Compound V3 RPC fetcher records `total_supplied_usd = 0.0` as a placeholder pending

wiring of the USDC base-units $\rightarrow$ USD conversion. Consequently `tvL_compound`, `debt_compound`, and the derived `dTVL_compound_24h` feature column are identically zero across all 13,105 hourly rows of `joined_clean.parquet`. Branch B of the forecaster therefore receives one constant input ($\approx 14\%$ of its 7-dimensional input capacity), which the model ignores after the first few epochs but which represents wasted capacity.

b) Ethereum gas price is a constant stub (Finding #4).: `data/fetch_gas_eth.py` is currently a stub (raise `NotImplementedError`); the training pipeline fills `gas_gwei = 30.0` as a constant for the entire data window. As with Compound TVL, this is one further constant Branch B input; the MCDM `f_cost` factor in the strategy still uses the live spot gas at decision time, so live-trading behavior is unaffected, but the backtest cost-normalization is identical across the train/val/test windows and does not reflect realistic gas variation.

c) Aggregate-only metrics, no per-quarter breakdown (Finding #9).: The headline metrics in `backtest/run_main.py` (net APY, annual Sharpe, max drawdown, Calmar, forecast hit-rate) are computed over the full test window as a single aggregate. Given the regime structure documented in `CLAUDE.md` (Q4 2024 standard deviation 5.75pp vs 2026 Q1 0.57pp — approximately $10\times$ volatility variation), this aggregation can mask edge concentration in a single regime. A per-quarter breakdown table is queued for the next paper cycle.

d) Composite-loss components logged in unscaled form (Finding #10).: `forecaster/losses.py` logs the bare MSE, $1 - \rho_w$, and quantile-loss components to MLflow rather than the scaled $\alpha \cdot \text{MSE}$, $\beta \cdot (1 - \rho_w)$, $\gamma \cdot Q_{90}$ contributions. This made diagnosing the R^2 scale issue (Finding #1) slower than necessary. Scaled-component logging is a low-risk addition deferred to a follow-up.

e) Tie-break bias in winner-series accounting (Finding #11).: `backtest/run_main.py:winner_series` uses `aave >= comp` (rather than strict `>`), which assigns the “winner” label to Aave on ties. In real data ties occur at machine precision only and the effect on `n_rebalances` is bounded by ± 1 ; we document this as a convention.

f) Boundary leakage from 24-hour pct_change feature (Finding #12).: `data/features.py` computes `dTVL_aave_24h = tvL_aave.pct_change(24)` on the full joined panel before the train/val/test split is applied, then slices by row index. The first 24 rows of each non-training fold therefore use TVL inputs from the previous fold. The contamination is bounded at $24/2501 \approx 0.96\%$ of test samples; we report it as a threat-to-validity rather than re-engineer the feature pipeline per-fold for this cycle.

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